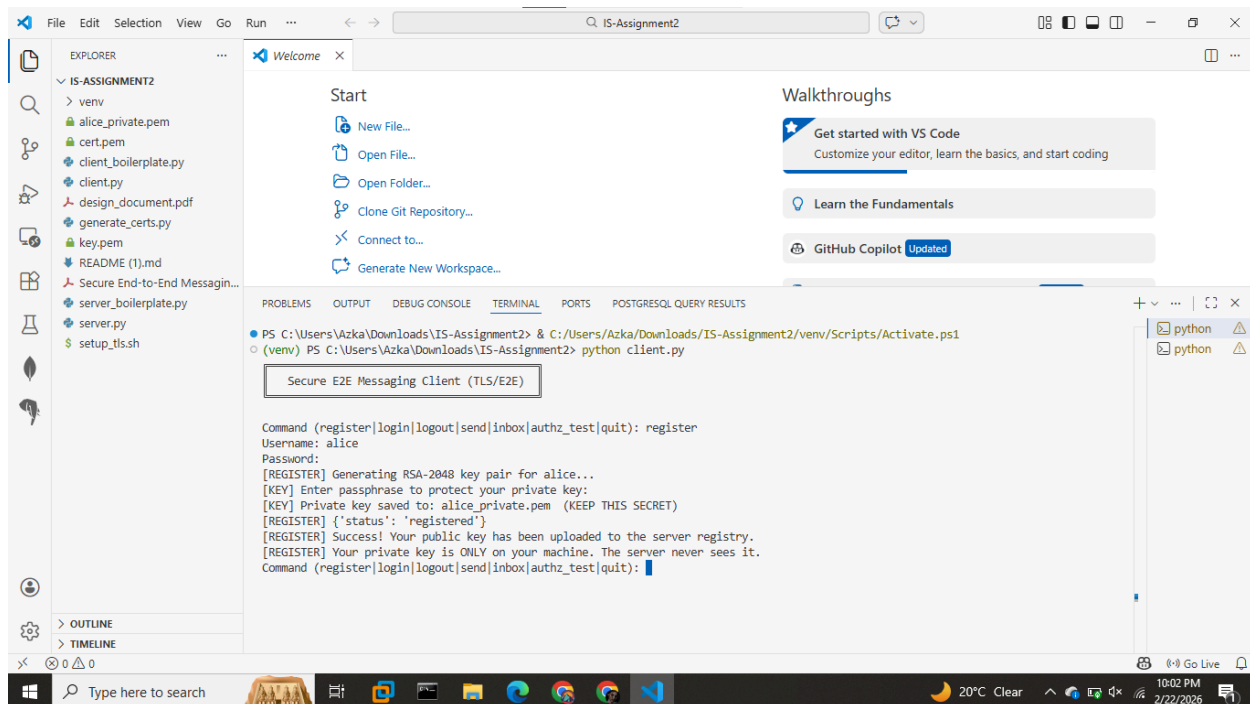


Server Starting (TLS proof)

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  POSTGRESQL QUERY RESULTS

• PS C:\Users\Azka\Downloads\IS-Assignment2> venv\Scripts\activate
>>
❖ (venv) PS C:\Users\Azka\Downloads\IS-Assignment2> python server.py
[SERVER] Starting secure HTTPS server on https://127.0.0.1:5443
[SERVER] Plain HTTP is NOT served. TLS only.
* Serving Flask app 'server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on https://127.0.0.1:5443
Press CTRL+C to quit
```

Alice Registers



```
File  Edit  Selection  View  Go  Run  ...  IS-Assignment2
EXPLORER
IS-ASSIGNMENT2
  venv
  alice_private.pem
  cert.pem
  client_boilerplate.py
  client.py
  design_document.pdf
  generate_certs.py
  key.pem
  README (1).md
  Secure End-to-End Messagin...
  server_boilerplate.py
  server.py
  setup_tls.sh

Start
  New File...
  Open File...
  Open Folder...
  Clone Git Repository...
  Connect to...
  Generate New Workspace...

Walkthroughs
  Get started with VS Code
  Learn the Fundamentals
  GitHub Copilot Updated

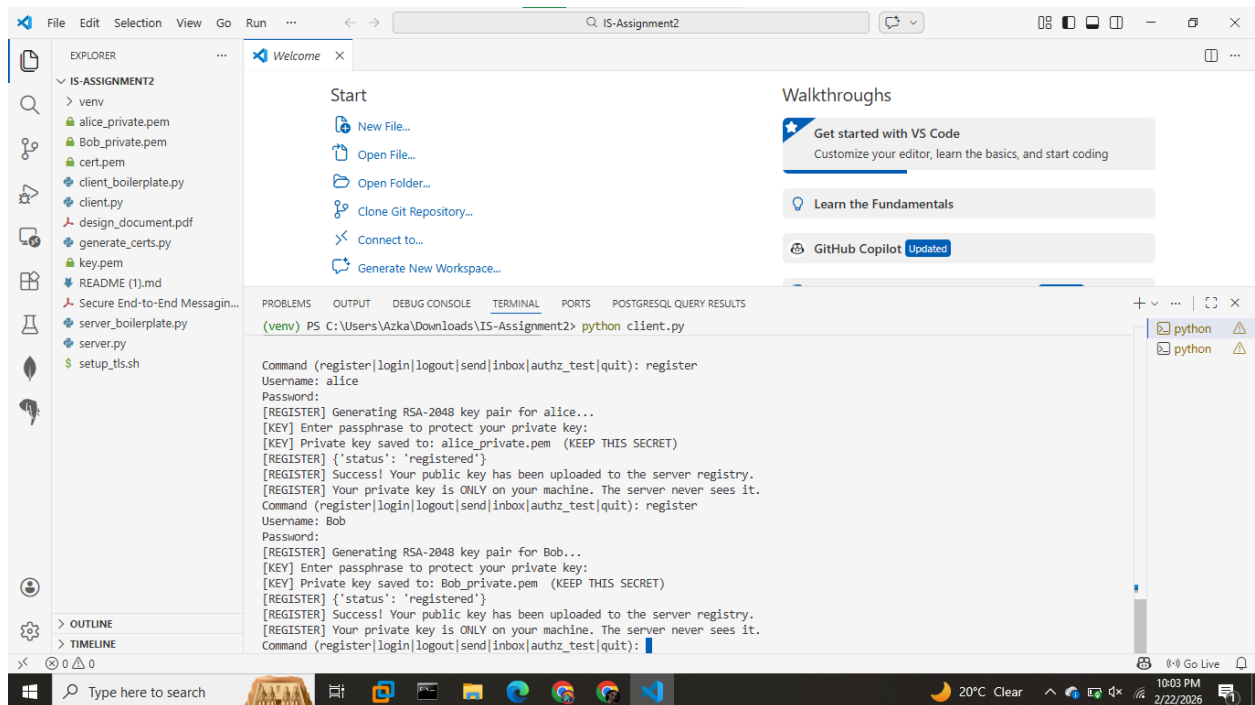
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  POSTGRESQL QUERY RESULTS

• PS C:\Users\Azka\Downloads\IS-Assignment2> & C:\Users\Azka\Downloads\IS-Assignment2\venv\Scripts\Activate.ps1
  (venv) PS C:\Users\Azka\Downloads\IS-Assignment2> python client.py

Secure E2E Messaging Client (TLS/E2E)

Command (register|login|logout|send|inbox|auth_test|quit): register
Username: alice
Password:
[REGISTER] Generating RSA-2048 key pair for alice...
[KEY] Enter passphrase to protect your private key:
[KEY] Private key saved to: alice_private.pem (KEEP THIS SECRET)
[REGISTER] {'status': 'registered'}
[REGISTER] Success! Your public key has been uploaded to the server registry.
[REGISTER] Your private key is ONLY on your machine. The server never sees it.
Command (register|login|logout|send|inbox|auth_test|quit):
```

Bob Registers



Alice Sends a Message

```
Command (register|login|logout|send|inbox|authz_test|quit): login
Username: alice
Password:
[KEY] Enter your private key passphrase:
[KEY] Private key loaded successfully.
[LOGIN] Authenticated as alice. Session token stored in memory.
Command (register|login|logout|send|inbox|authz_test|quit): send
To (username): Bob
Message: Hello Bob, I am Alice
[SEND] Fetched Bob's public key from server registry.
[SEND] Message encrypted with AES-256-GCM.
[SEND] AES key encrypted with RSA-OAEP (recipient's public key).
[SEND] {'status': 'message stored'}
Command (register|login|logout|send|inbox|authz_test|quit):
```

Server Logs

```

❖ (venv) PS C:\Users\Azka\Downloads\IS-Assignment2> python server.py
[SERVER] Starting secure HTTPS server on https://127.0.0.1:5443
[SERVER] Plain HTTP is NOT served. TLS only.
* Serving Flask app 'server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on https://127.0.0.1:5443
Press CTRL+C to quit
127.0.0.1 - - [22/Feb/2026 22:02:15] "POST /register HTTP/1.1" 200 -
127.0.0.1 - - [22/Feb/2026 22:03:03] "POST /register HTTP/1.1" 200 -
127.0.0.1 - - [22/Feb/2026 22:05:04] "POST /login HTTP/1.1" 200 -
127.0.0.1 - - [22/Feb/2026 22:07:21] "GET /public_key/Bob HTTP/1.1" 200 -
127.0.0.1 - - [22/Feb/2026 22:07:21] "POST /send HTTP/1.1" 200 -

```

Bob Decrypts

```

Command (register|login|logout|send|inbox|authz_test|quit): login
Username: Bob
Password:
[KEY] Enter your private key passphrase:
[KEY] Private key loaded successfully.
[LOGIN] Authenticated as Bob. Session token stored in memory.
Command (register|login|logout|send|inbox|authz_test|quit): inbox

[INBOX] 1 message(s):
=====
From:    alice
Message: Hello Bob, I am Alice
-----

Command (register|login|logout|send|inbox|authz_test|quit): 

```

Authorization Failure

```

Command (register|login|logout|send|inbox|authz_test|quit): authz_test
Try to access inbox of user: alice

[AUTHZ TEST] Trying to access alice's inbox as Bob:
HTTP 403: {'error': 'Forbidden - you can only read your own messages'}
Command (register|login|logout|send|inbox|authz_test|quit): 

```